Candidate Number……....

THE PRESIDENT'S OFFICE – RALG

TAHOSSA ROMBO DISTRICT EXAMINATION SYNDICATE

FORM FOUR MOCK EXAMINATION

032/2A **CHEMISTRY 2A (ACTUAL PARCTICAL)**

TIME: 2:30 Hours **August, 2024**

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Instructions

1. This paper consists of **two (2)** questions. Answer all questions.
2. Question 1 carry **30 marks** and Question 2 carry **20 marks**
3. Qualitative analysis guide sheet authorized by NECTA and non-programmable calculators may be used.
4. **Cellular phones and any unauthorized materials** are not allowed in the examination room.
5. Write your **examination Number** on every page of your answer sheet(s).
6. You may use the following constants:
   Atomic masses: H=1, C=12, O=16, Na=23, Cl=35.5,S=32
   1 litre =1 dm^3=1000 cm^3

FOR ASSESSOR'S ONLY		
QUESTION NUMBER	**SCORE**	**MARKER'S INITIALS**
1		
2		
TOTAL		
CHEKER'S INITIALS		

1. You are provided with the following:
   i. **Solution D** – Containing 5.3g of T_2CO_3 per 0.50 dm^3 of solution.
   ii. **Solution N** – containing 1.55g of hydrochloric acid per 200cm^3 of solution.
   iii. **Methyl orange Indicator** solution

PROCEDURE:

Put the acid solution in the burette. Pipette 20 cm3 (or 25 cm3) of **D** into a titration flask. Add few drops of Methyl orange indicator. Titrate solution **D** against the acid solution until an endpoint is reached. Record your titration results in the table shown below. Repeat the titration procedure until you obtain three or four other titre values.

a) Table of results (burette readings)

Titration	Pilot	1	2	3
Final burette reading (cm^3)				
Initial burette reading (cm^3)				
Volume used.				

The volume of the pipette used was……cm^3.
The volume of the burette used was…….cm^3.

Summary:

……..cm^3 of solution **N** was used ……..cm^3 of solution **D** for complete Neutralization.

b) The colour change at the end-point was from…….to…….

c) The volume of acid solution for the above neutralization reaction was…….cm^3

d) The balanced equation for the above neutralization reaction is:

e) Calculate:

i. The molarity of solution **N**
ii. The molarity of solution **D**.
iii. The molecular weight of T_2CO_3
iv. The atomic mass of element **T**.
v. The name of Element **T** is _____

2. You are provided with the following materials

- **TT**: A solution of 0.13M $Na_2S_2O_3$ (Sodium thiosulphate)
- **HH**: A solution of 2MHC1;
- Distilled water:
- Stopwatch
- Beaker 100ml
- Piece of paper marked **X**

PROCEDURE:

i). Using $10cm^3$ measuring cylinder. Measure $20cm^3$ of solution **TT** and put into 100 cm^3 beaker.
ii). Use different measuring cylinder to measure $10cm^3$ of **HH** and pour it into the beaker containing solution **TT**.
Immediately start the stop watch. Swirl the beaker twice.
iii). Place the beaker with contents on top of a piece of paper marked **X**
iv). Look down vertically through the month of the beaker so as to see the cross at the bottom of the beaker. Stop the clock when the letter **X** is **invisible.**
v). Record the time taken for the letter **X** to disappear completely.
vi). Repeat the experiment as shown in table below.
vii). Record your results in tabular form as shown in table below.

Experiment No	**Volume of HH (cm^3)**	**Volume of TT (cm^3)**	**Volume of Distilled Water (cm^3)**	**Time in (sec)**	**Rate(1/t) (sec^{-1})**
1	10	20	0		
2	10	15	5		
3	10	10	10		
4	10	5	15		

Questions:

a) Complete filling the table of results in the table above

b) Write a balanced equation for reaction between **TT** and **HH**

c) What is the reaction product which causes the solution to could the letter **X**?

d) How was the factor of concentration varied in this experiment?

e) Plot a graph of 1/t against the volume of the thiosulphate.

f) Use the graph to explain how variation of concentration affects the rate of chemical reaction